

BTS

Report AV2025003
October 23, 2024

The Bureau of Transportation Statistics Verifies the Accuracy of Flight Delay and Cancellation Data but Can Do More To Assess Its Completeness and Consistency

Highlights

Bureau of Transportation Statistics (BTS)
AV2025003 | October 23, 2024

The Bureau of Transportation Statistics Verifies the Accuracy of Flight Delay and Cancellation Data but Can Do More To Assess Its Completeness and Consistency

Self-Initiated

Our Objective(s)

To assess the controls BTS has in place to ensure the completeness and accuracy of data for on-time performance and reported causes of delays and cancellations.

Why This Audit?

Flight disruptions, including both flight delays and cancellations, are a part of many passengers' experiences. Air carriers are required to submit to BTS on-time performance data—such as departure and arrival times, elapsed flight times, and minutes of delay—as well as the reported causes of any delayed or canceled flights. Accurate reporting of these data is important for the Department of Transportation (DOT) to understand the causes of delays and cancellations and take actions to protect consumers.

What We Found

BTS relies on informal procedures to verify the accuracy of on-time performance data and lacks procedures to verify the data's completeness.

- BTS has established informal quality control procedures to identify inaccuracies in on-time performance data submitted by reporting carriers. For example, BTS developed an automated process that performs 117 automated checks of the data to identify any errors or inconsistencies.
- However, BTS has not yet formally adopted these procedures nor established a plan to ensure their continued execution.
- Although BTS requires reporting carriers to certify that the on-time performance information they submit is correct and complete, the Agency does not have a method for verifying that carriers are reporting data for all flights, raising concerns about the data's reliability.

BTS lacks effective guidance and procedures for verifying reported causes data.

- BTS' technical directives, which provide guidance to carriers on reporting the causes of flight delays and cancellations, lack clarity, leading air carriers to interpret the guidance and report their data inconsistently.
- BTS lacks procedures to identify and address inconsistencies in reported causes data. Instead, the Agency relies on methods the air carriers are using to ensure their submitted data is accurate.
- We found a high rate of disagreement between data from FAA on reported causes versus the data that carriers report to BTS. BTS has not taken steps to understand and address the causes of these discrepancies.



7 **Recommendations** to improve the completeness and consistency of on-time performance and reported causes data. *(p. 16)*

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U.S. Department of Transportation
Office of Inspector General

Memorandum

Date: October 23, 2024

Subject: ACTION: The Bureau of Transportation Statistics Verifies the Accuracy of Flight Delay and Cancellation Data but Can Do More To Assess Its Completeness and Consistency | Report No. AV2025003

From: Nelda Z. Smith
Assistant Inspector General for Aviation Audits

To: Director, Bureau of Transportation Statistics

Across the National Airspace System (NAS), air carriers operate thousands of scheduled flights, transporting millions of passengers each day.[1] Flight disruptions, including both flight delays[2] and cancellations,[3] are a part of many passengers' experiences. Among the oversight responsibilities of the Department of Transportation (DOT) for the NAS is the function of gathering data on the timeliness of airline operations. Specifically, DOT analyzes systemwide and carrier-specific delay and cancellation data collected by its Bureau of Transportation Statistics' (BTS)[4] Office of Airline Information (OAI)[5] to understand the nature and causes of flight delays and cancellations.

Federal regulations[6] require reporting carriers[7] to collect and report the causes of delays and cancellations to BTS. The data that reporting carriers submit to BTS include the carrier's on-time performance data—such as departure and arrival

[1] Based on OIG's analysis of the January 2022 through December 2022 Air Traffic Consumer Reports.
[2] A flight is considered delayed when it is 15 minutes later than its scheduled arrival time.
[3] A flight is considered canceled when it is not operated even though it was listed in a carrier's computer reservation system within 7 calendar days of its scheduled departure.
[4] Part of DOT, BTS compiles, analyzes, and publishes information on the Nation's transportation system, including on commercial aviation.
[5] OAI collects, processes, and disseminates aviation data including airline reported flight delays and cancellation data. OAI is fully funded by the Federal Aviation Administration (FAA) through an Interagency Agreement.
[6] 14 Code of Federal Regulations (C.F.R.) Part 234.
[7] A reporting carrier is any airline that accounts for 0.5 percent or more of domestic scheduled passenger revenues and other air carriers that choose to report data voluntarily. Reporting carriers are required to submit on-time performance data each month to BTS' Office of Airline Information. In 2022, there were 17 reporting carriers. Prior to 2018, the reporting threshold was 1 percent or more.

times, elapsed flight times, and minutes of delay—as well as the reported causes of any delayed or canceled flights.

DOT's responsibility in gathering this data is related to another of the Department's oversight responsibilities for the NAS. DOT's Office of Aviation Consumer Protection (OACP)[8] plays an important role in safeguarding the rights and interests of air travelers. OACP uses the data from BTS as one of several tools[9] to monitor and understand the nature and causes of flight delays and cancellations. OACP can take actions to enforce aviation consumer protections when warranted. To those ends, accurate reporting of on-time performance and reported causes data is important for the Department to gain an understanding of the causes of delays and cancellations. The reported causes data can also assist OACP in taking actions to protect consumers. Accordingly, we initiated our audit to assess the controls BTS has in place to ensure the completeness and accuracy of data for on-time performance and reported causes of delays and cancellations.

We conducted our work in accordance with generally accepted Government auditing standards. Exhibit A details our scope and methodology. Exhibit B lists the organizations we visited or contacted, and exhibit C lists the acronyms used in this report.

We appreciate the courtesies and cooperation of DOT representatives during this audit. If you have any questions concerning this report, please contact me or Jay Borwankar, Program Director.

cc: The Secretary
DOT Audit Liaison, M-1
FAA Audit Liaison, AAE-001

[8] DOT's OACP is a unit within the Office of General Counsel that, among other things, is responsible for monitoring and enforcing aviation consumer protections including those stemming from unrealistic or deceptive scheduling such as from chronically delayed flights.

[9] OACP also monitors news reports using automated alerts to identify cancellation and delay issues and their causes. OACP also obtains information on delayed flights and causes from the reporting carriers.

Background

BTS is one of 13 principal Federal statistical agencies and is the source of statistics on commercial aviation, multimodal freight activity, and transportation economics. Its work provides context to decision makers and the public for understanding statistics on transportation. BTS' mission is to assure the credibility of its products and services through rigorous analysis, transparent data quality, and independence from political influence. BTS aims to promote innovative methods of data collection, analysis, visualization, and dissemination to improve operational efficiency, examine emerging topics, and create relevant and timely information products that foster understanding of transportation and its transformational role in society. As a principal Federal statistical agency, BTS is governed primarily by statute[10] and statistical policy directives of the Office of Management and Budget.

In November 2002, DOT issued its regulation requiring reporting carriers to collect and report the causes of flight delays and cancellations to BTS. The reported causes for flight delays are categorized in five broad categories:

- **Air carrier (Airline):** The cause of the cancellation or delay was due to circumstances within the airline's control (e.g., lack of flight crew, maintenance, aircraft cleaning, baggage loading, fueling, etc.).
- **Extreme weather:** Significant meteorological conditions (actual or forecasted) that, in the judgment of the carrier, delays or prevents the operation of a flight such as thunderstorm, deicing aircraft, tornado, hurricane, or snowstorm.
- **NAS:** Delays and cancellations attributable to the national aviation system that refer to a broad set of conditions, such as non-extreme weather conditions[11], airport operations, heavy traffic volume, and air traffic control.
- **Late-arriving aircraft:** A previous flight with the same aircraft arrived late, causing the present flight to depart late.
- **Security:** Delays or cancellations caused by evacuation of a terminal or concourse, reboarding of aircraft because of security breach, inoperative

[10] 49 U.S. Code (U.S.C.) § 6302.
[11] In the November 2002 regulation, DOT provided additional discussion of the difference between "extreme" and "non-extreme" weather conditions for purposes of reporting the causes of flight delays and cancellations.

screening equipment, and/or long lines in excess of 29 minutes at screening areas.

The categories for the reported causes of flight cancellations are the same, except there is no category for late-arriving aircraft. Reporting carriers are only required to report the causes of delays when flights experience an arrival delay of 15 minutes or more, and they are only required to report each cause of delay that contributes 5 minutes or more to the delay. BTS publishes the flight delay and cancellation data in its Airline Service Quality Performance (ASQP) database,[12] which is available publicly on the Agency's website. The Federal Aviation Administration (FAA) also collects and reports flight delay and cancellation data for all domestic flight operations through its Aviation System Performance Metrics (ASPM)[13] database. However, FAA is only required to collect data for all flights that experience NAS delays of 15 minutes or more. FAA uses the data it collects to analyze the NAS' performance, as well as to make strategic decisions regarding NAS management.

Annually, BTS issues a technical directive that identifies all reporting carriers for the upcoming reporting cycle[14] and reinforces all applicable data requirements. The technical directive also provides guidance on how reporting carriers should classify and report their activity. BTS' data is used by carriers to benchmark their operational performance against rival carriers, and by academia and other industry stakeholders to analyze flight delays and cancellations.

Separately, OACP uses BTS' flight delay and cancellation data to monitor and analyze airline operating and scheduling performance. OACP uses these data for three primary purposes:

1. Every month OACP uses the data it receives from BTS to compile its Air Travel Consumer Reports (ATCR)[15] publication. Among other things, the ATCR provides data on the on-time performance of reporting carriers and the causes of flight delays.

[12] ASQP is a publicly available database that captures information about on-time-performance, delays and cancellations, and their reported causes information.
[13] ASPM provides data on flights to and from 77 airports that have been designated as ASPM airports, and includes flights operated by reporting carriers. Among other things, ASPM captures a flight's scheduled and actual gate departure time, runway departure time, runway arrival time, and gate arrival time.
[14] BTS' reporting cycle begins in January and ends in December.
[15] ATCR, which OACP publishes monthly, is designed to assist consumers with information on the quality of services provided by the airlines. The portions of the report based on data collected by BTS address: Flight Delays/Cancellations, Mishandled Baggage, Wheelchairs and Scooters, and Oversales.

2. OACP conducts regular meetings with reporting carriers to go over their operating performance and to understand the factors underlying their flight delays and cancellations.

3. OACP uses on-time performance data to identify potential consumer protection violations such as those stemming from chronically delayed flights or other unrealistic scheduling practices. When there are potential violations, OACP staff refer to BTS causal data for a general understanding of the situation, and would obtain more granular information on the cause of delay from reporting carriers.

BTS Relies on Informal Procedures To Verify the Accuracy of On-Time Performance Data and Lacks Procedures To Verify the Data's Completeness

BTS has established informal quality control procedures to identify inaccuracies in on-time performance data submitted by reporting carriers. However, BTS has not yet formally adopted these procedures. Moreover, BTS has not established a plan to ensure the continued execution of its quality assurance procedures. Finally, BTS does not have a method for verifying that carriers are reporting data for all flights.

BTS Has Procedures To Verify the Accuracy of On-Time Performance Data, but It Has Not Formally Adopted Them

BTS has developed quality control procedures that allow it to identify and resolve inaccuracies in the on-time performance data it receives from reporting carriers. However, the Agency has not formally adopted these procedures.

BTS Has Developed Quality Control Procedures To Identify and Resolve Inaccuracies in Carrier-Reported On-Time Performance Data

BTS relies on manual and automated quality control procedures that allow the Agency to identify and resolve inaccuracies in the on-time performance data it receives from reporting carriers. DOT's regulation requires reporting carriers to submit their on-time performance data to BTS within 15 days following the end of the month. BTS requires that carriers use the format prescribed in its Form 234,

entitled the On-Time Flight Performance Report,[16] to submit data via the e-Submit[17] web application. Reporting carriers also use Form 234 to concurrently submit their reported causes data for all delayed and canceled flights. In fulfilling BTS' Form 234 reporting requirements, the six carriers in our sample told us they use automated systems for collecting their flight activity data. Among other things, automated systems collect data about flight departure and arrival times.[18]

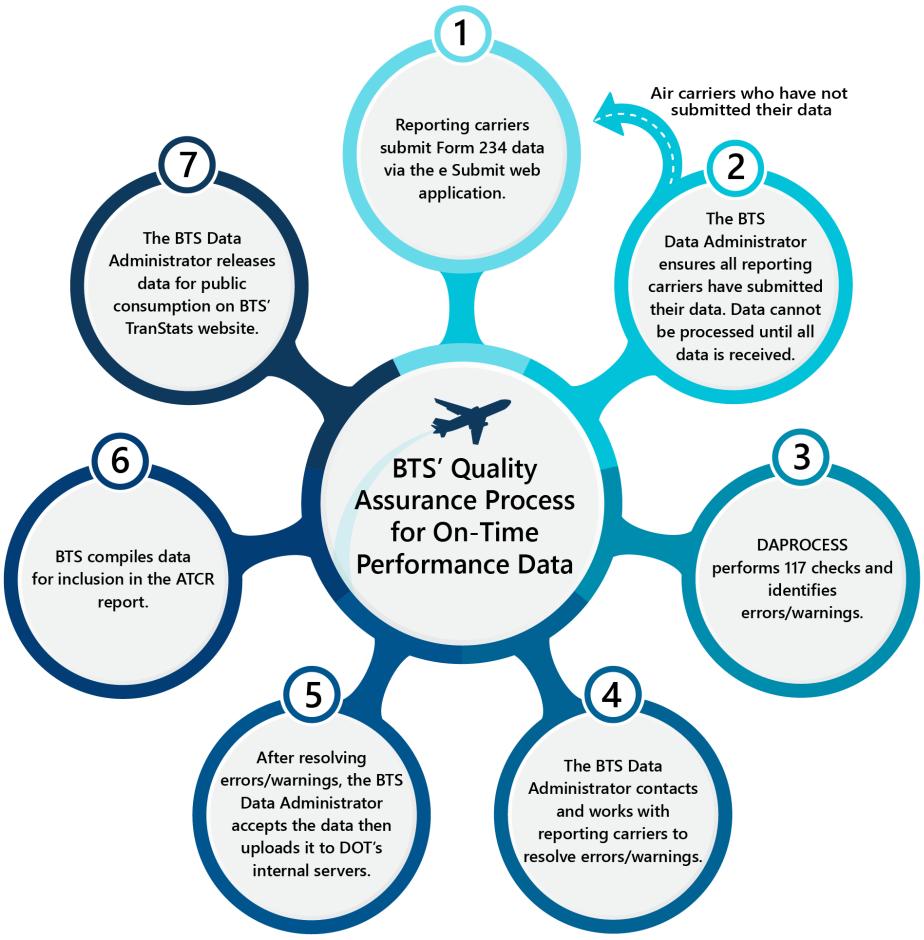
BTS also requires that reporting carriers' monthly submissions include a transmittal letter, which among other things includes a certification statement to BTS. In the certification statement, the carriers certify that the Form 234 information they are submitting is true, correct, and complete to the best of their knowledge. Along with carriers' certification statement, BTS has also developed manual and automated quality control procedures for processing the on-time performance data it receives every month. The Agency currently relies on one staff member—the BTS Data Administrator—to execute these quality control procedures. The BTS Data Administrator is responsible for executing a number of manual procedures every month as part of the Agency's quality control process. For example, the BTS Data Administrator performs a cross-check of the reporting carriers' names listed on that year's technical directive to the carriers who have submitted Form 234 data to make sure that all reporting carriers have submitted their on-time performance data.

In addition, BTS has developed an automated process called DAPROCESS, which performs 117 automated checks of the data and identifies any errors or inconsistencies in the data. Based on our review, the most common types of errors identified through DAPROCESS are due to miscalculation of the minutes of delays based on a flight's departure and arrival times. Once BTS processes all reporting carriers' Form 234 records through DAPROCESS, the BTS Data Administrator is responsible for contacting the carriers and working with them to resolve all errors/warnings the system identifies. According to the BTS Data Administrator, the additional processing of the data cannot move forward until all errors/warnings have been clarified and/or corrected. Next, the BTS Data Administrator uploads the data to DOT's internal servers and makes the data available to other OAI staff members for further analysis. The BTS Data

[16] Reporting carriers are required to report data for flights involving any airport in the 48 contiguous states. Among other things, they have to report carrier and flight numbers, aircraft tail numbers, origin and destination airport codes, scheduled departure and arrival times, actual departure and arrival times, and minutes of delays.
[17] BTS' e-Submit web application is designed to allow reporting air carriers to submit airline data via the internet. Data submitted includes but is not limited to the reporting carrier's identification code, flight numbers, scheduled arrival/departure times, gate departure/arrival times, and any applicable minutes of delay or canceled flights.
[18] Each of the six carriers we interviewed said they use the Aircraft Communication Addressing and Reporting System (ACARS) to collect their departure and arrival data. ACARS is a digital datalink system used for transmitting short messages between aircraft and ground stations. ACARS is used for various purposes such as sending and receiving flight plans, weather reports, and aircraft performance data.

Administrator also organizes and compiles on-time performance and reported causes data for use in OACP's monthly Air Traffic Consumer Report (ATCR). Finally, the BTS Data Administrator releases the data for public consumption on the Agency's TranStats[19] website, and OAI provides the BTS data to OACP in the form of tables and airline rankings for inclusion in OACP's monthly ATCR report. The figure below provides an overview of the manual and automated processes BTS uses to collect, validate, and disseminate on-time performance and reported causes data.

Overview of BTS' Quality Assurance Process for Collecting, Validating, and Disseminating On-Time Performance and Reported Causes Data



Source: OIG analysis of BTS' process

[19] TranStats is a website for transportation researchers and analysts aimed at providing "one-stop shopping" for transportation data.

BTS Has Not Formally Adopted Its Procedures for Verifying Data Accuracy

Although BTS has a process for identifying inaccuracies in on-time performance data, as described above, the Agency has not yet formally adopted these procedures. In May 2023, after OIG initiated its audit, BTS developed draft guidance that documents the Agency's current quality assurance process for collecting, validating, and disseminating airline-reported Form 234 data. However, since that time, the Agency has not updated its draft standard operating procedures (SOPs)[20] to reflect this guidance or formally adopted the SOPs. According to BTS officials, even though the guidance has not been formally adopted, the Agency is following the quality assurance process, and since the draft quality assurance processes serve as internal guidance only, they have not been officially adopted.

After our review of the 2023 draft operating procedures, a senior OAI official provided SOPs BTS developed in 2009 for our review to demonstrate the Agency had longstanding policies in place. However, these SOPs are inadequate. First, the SOPs are outdated and do not cover the Agency's current quality assurance process for collecting, processing, reviewing, and reporting airline reported Form 234 data. Second, the SOPs are in draft and have never been formally adopted by the Agency. Finally, the existence of these SOPs was not common knowledge within the Agency, as reflected by the fact that the BTS staff we interacted with throughout the audit, including the BTS Data Administrator, were not aware of the 2009 SOPs' existence.

The Federal statute that established BTS states that the Director of BTS shall encourage the standardization of data, data collection methods, and data management and storage technologies for data collected by the Agency. Also, BTS' Statistical Standards Manual (statistical manual)[21] states that data processing procedures must be documented, and that the Agency's data may be subjected to an external peer review, which would require the Agency to have documented procedures. Additionally, the manual establishes professional standards to ensure consistency among studies conducted by statistical agencies to provide users clear documentation of the methods used to collect, process, analyze, and disseminate statistical information.

Without having adopted formal written procedures, BTS runs the risk of procedures not being consistently followed or properly executed, particularly if the Agency must rely on new employees or personnel unfamiliar with the quality assurance process. Additionally, the lack of formal documented procedures limits

[20] BTS' Office of Airline Information Draft Standard Operating Procedures (updated on July 27, 2009).
[21] BTS Statistical Standards Manual (October 2005).

the ability of public data users and peer reviewers to understand how BTS reports the data.

BTS Has Not Established a Plan To Ensure the Continued Execution of Its Quality Assurance Procedures

BTS currently relies on one staff member to complete quality assurance procedures. Beginning in December 2023, the BTS Data Administrator had to assume some additional responsibilities following the retirement of another BTS staff member. For example, once the Form 234 data has been processed through the Agency's quality assurance procedures, the BTS Data Administrator is now responsible for performing a rolling 13-month trend analysis of the data. Among other things, the trend analysis identifies material changes in a reporting carrier's on-time and flight cancellation performance.[22] According to a senior BTS official, the Agency does not currently intend to backfill for the retired staff member's position due to budget constraints that will likely extend through the third quarter of fiscal year 2025.

Moreover, BTS has not established a succession plan to ensure coverage of these quality control tasks if the BTS Data Administrator were to leave the Agency or require an extended absence. DOT's Guide to Workforce Planning[23] requires DOT Operating Administrations to identify among other things the need for succession planning—i.e., to develop a replacement plan for specific mission critical positions. The Federal statute[24] that established BTS states that the Director of BTS shall continually improve data collection methods to improve the accuracy and utility of transportation statistics. This position performs an important role in ensuring the reliability and accuracy of the data and is closely aligned with BTS' mission. The Agency has identified mission-critical positions to include leadership positions such as BTS executives, supervisory office directors, and managers for individual BTS products and services. However, the Agency has not undertaken an assessment to identify whether the BTS Data Administrator position and other non-leadership positions within the organization are mission-critical and has not developed a succession plan for them. Without a succession plan to cover the BTS Data Administrator position's responsibilities,

[22] On a rolling 13-month basis, BTS generates a trend analysis of reporting carriers' on-time and cancellation performance. If there are variances of 5 percent or more for delays and 2 percent or more for cancellations from the previous month to the current month, the BTS Data Administrator will highlight these variances for the Agency's own internal awareness.
[23] U.S. Department of Transportation Human Capital Management Guide to Workforce Planning (Revised June 2008).
[24] 49 U.S.C. § 6302(b)(3)(B)(v).

BTS' ability may be limited to ensure the continuity of operations in a critical function and thus safeguard the accuracy and reliability of data it disseminates.

BTS Does Not Verify That Carriers Report Data for All Flights

BTS lacks controls to verify the completeness of the on-time performance data submitted by carriers. Specifically, BTS requires reporting carriers to certify that the Form 234 information they are submitting is true, correct, and complete to the best of their knowledge. However, until BTS establishes a quality control process for assessing that carriers are submitting data for all flights, the Agency cannot be certain that the data the carriers are submitting is complete. Establishing quality control procedures is also in alignment with BTS' stated goal of improving the accuracy and utility of transportation statistics particularly given that the data is disseminated to the public.[25] DOT's Information Dissemination Quality Guidelines[26] further emphasize, among other things, the importance of data disseminated by DOT to be accurate and complete.

BTS has considered multiple options for additional quality control procedures to verify the data's completeness, according to a BTS official. One BTS official told us that the Agency could compare the Form 234 data it receives from reporting carriers to other data sources, such as BTS' T-100[27] dataset and/or the Official Airline Guide (OAG)[28] dataset, to validate the data's completeness. However, according to this BTS official, since 2016, the Agency's staffing has shrunk from 12 Federal full-time employees (FTEs) and 12 contractors to 6 Federal FTEs and 5 contractors (including 3 full-time and 2 part-time) in 2024. According to senior OAI officials, these staffing reductions limit BTS' ability to perform additional analyses.

While we support BTS' consideration of additional quality control options, with regard to comparing the Form 234 data to the data from BTS' T-100 and OAG, we note that these datasets also report data submitted by the air carriers. Therefore, a verification based on this comparison alone may not assist BTS fully in independently assessing the air carrier data's completeness. According to senior BTS officials, the Agency has discussed the benefits of performing such analysis

[25] 49 U.S.C. § 6302(b)(3)(B)(iv).
[26] DOT-OST-2019-0135.
[27] The T-100 traffic reports contain domestic market data reported by both U.S. and foreign air carriers, including carrier, origin, destination, and service class for enplaned passengers, freight, and mail when both origin and destination airports are located within the boundaries of the United States and its territories.
[28] OAG is the industry's leading airline data provider. OAG provides data on airline schedules, cargo, and aviation analytics.

and is exploring alternative data sources to validate the completeness of the data as resources permit. Regardless, the lack of a quality control process for ensuring the completeness of reporting carriers' on-time performance data raises concerns about the data's reliability.

BTS Lacks Effective Guidance and Procedures for Verifying Reported Causes Data

BTS' technical directives, which provide guidance to carriers on reporting the causes of flight delays and cancellations, lack clarity, resulting in carriers interpreting the guidance and reporting their data inconsistently. Furthermore, BTS lacks procedures to identify and address inconsistencies in reported causes data.

BTS' Unclear Guidance Results in Airlines Reporting Cause Data Inconsistently

DOT's regulation requires reporting carriers to collect and report the causes of airline delays and cancellations. BTS' technical directives, which are issued annually, provide guidance to reporting carriers on how they should collect and report their Form 234 data to provide a better understanding of the causes of delays and cancellations. However, our review has found that unclear guidance in BTS' technical directives has resulted in carriers' interpreting the guidance inconsistently. As a result, some carriers submit reporting cause data differently than others, which introduces inconsistencies into the data reported to BTS.

Specifically, in the same technical directive, BTS states that carriers have the option of reporting multiple causes of departure delays or the predominant cause of departure delays. However, the technical directive does not sufficiently clarify that a carrier can only choose the predominant cause to report departure delays "when a delay cause has fewer than 6 minutes of delays." DOT's regulation states that "reporting carriers are required to code delays only when the arrival delay is 15 minutes or greater; and reporting carriers must report each causal component of the reportable delay when the causal component is 5 minutes or greater." While BTS' technical directive[29] does reiterate this regulatory requirement, it further adds to inconsistent reporting because the directive asks

[29] BTS' technical directive #27, issued on August 15, 2017.

that carriers account for departure delays of 6 minutes or longer, instead of the regulation's requirement to report departure delays of 5 minutes or longer.

BTS' unclear guidance has resulted in some carriers submitting their reporting cause data differently than others. For example, some carriers are grouping and reporting all minutes of delays under one predominant cause. Furthermore, because BTS' guidance does not define what the Agency means by predominant cause or how carriers should use it to report their data, carriers have used their judgement differently in interpreting how and when to report a predominant cause. One of the six carriers we spoke with told us that that they interpreted the predominant cause to mean the delay cause with the longest duration. Another carrier told us that they interpreted the predominant cause to mean the root cause of the delay. In all, the six airlines in our sample are using three different methodologies in providing their reported causes data to BTS: (1) two carriers use multiple causes, (2) three use predominant cause by longest duration, and (3) one uses predominant cause by root cause.

Effectively, the technical directive allows for multiple interpretations of delay events. By allowing multiple interpretations, the guidance creates opportunities for an inconsistent and incomplete understanding of the causes of delays and cancellations. Moreover, these data inconsistencies make it difficult to compare the operational performance of one carrier to another carrier who may have used a different methodology to report their data. In one instance, we came across a carrier who switched from using the multiple causes approach to report their data to the predominant cause approach. In this case, the resulting inconsistency in the data would make it difficult to compare that carrier's operating performance from before they switched their reporting methodology to after they switched. Therefore, any analysis of inconsistent data could lead to an unreliable assessment of problems as well as the actions taken to address them.

According to several BTS officials, the reason for the wording in the technical directives being unclear is unknown. The officials agreed that the lack of clarity results in inconsistencies in the data collected and distributed by BTS, which raises concerns about the data's reliability.

BTS Lacks Procedures To Identify and Address Inconsistencies in Reported Causes Data

The certification statement that reporting carriers submit with their Form 234 data each month also covers reported causes data. However, outside of this requirement, BTS has not developed any other quality control procedures to verify the consistency of carrier-submitted reported causes data. As a result, the

consistency of BTS' data on the causes of flight delays and cancellations rests entirely on how accurately the carriers are reporting their data.

Because BTS has not developed procedures to verify reported causes data, the Agency relies on methods the air carriers are using to ensure their submitted data is accurate. The six randomly selected carriers we spoke with told us that they have adopted their own quality control procedures for collecting and categorizing their reported causes data. These carriers told us that they collect their reported causes data at a much more granular level than the four or five broad categories that BTS provides for reporting this data. For example, 2 of the 6 carriers told us they use over 100 delay codes to capture the causes of delays and cancellations. All air carriers we interviewed told us that having access to granular data on delays and cancellations allows them to better understand the inefficiencies in their operations and develop a more targeted response to alleviate those inefficiencies. Additionally, the carriers generally assign the responsibility for managing different causal categories to different individuals or groups, which creates an incentive to ensure that any inconsistencies in assigning delay codes are identified and addressed. All six carriers we spoke with have also established a process to research and arbitrate any disagreements between departments. In some cases, carriers have also established an internal audit function to ensure that the carrier is capturing delay or cancellation causes under the appropriate categories. The carriers we spoke with told us that they have also developed custom methodologies to consolidate the granular data into the four or five categories that are captured in their Form 234 data and reported to BTS.

Based on our discussions with these six carriers, we were able to determine that they were using three different methodologies to report their causal data, resulting in inconsistencies in the data. However, BTS has not recognized the inconsistencies between the carriers' approaches as an issue, and therefore has not developed procedures to identify and reconcile such inconsistencies.

Moreover, our analysis identified another area of inconsistency in BTS' data on reported causes. Specifically, FAA collects reported causes data from the air traffic control system. FAA then compares the reported causes data it collects for all delayed flights that are attributed to the NAS with the data that airlines report to BTS. This analysis identifies "no-matches"—i.e., instances where FAA and airlines (as reflected in BTS' data) disagree on specific aspects of the NAS being the reported cause of a delay.[30] We conducted our own no-matches analysis and found a high rate of disagreement between FAA's data and BTS' data as it pertains to NAS-caused delays and cancellations between 2018 to 2022 (see table). The table below also highlights that the rate of no-matches for

[30] "No Match" is assigned to flights where FAA and airlines (in their reports to BTS) disagree on whether the reported minutes of delay attributed to the NAS match each other, either the assigned minutes or underlying causes.

NAS-attributed delayed and canceled flights has increased over the past few years.

Rate of No-Matches Between FAA's and BTS' NAS-Attributed Delay Data for Delayed and Canceled Flights

Year	Rate of NAS-Attributed Delays[31]	Rate of Delay No-Matches in FAA's Data	Rate of NAS-Attributed Cancellations	Rate of Cancellation No-Matches in FAA's Data
2018	55.0%	24.0%	21.4%	12.0%
2019	54.1%	22.0%	19.5%	9.0%
2020	52.0%	41.6%	1.6%	36.0%
2021	43.9%	42.8%	7.8%	25.4%
2022	45.3%	42.4%	8.4%	24.6%

Source: OIG analysis of FAA's ASPM and ASQP data

Note: The rates of NAS-attributed delays and cancellations are calculated as the number of delayed and canceled flights attributed to the NAS divided by the total number of delayed and canceled flights. The rate of no-matches for delayed and canceled flights is calculated as the number of NAS-delayed or -canceled flights that were classified as no-matches divided by the total number of NAS-delayed or -canceled flights.

According to FAA officials, some of the no-matches can be explained by methodological differences in how the data is collected. First, BTS data comes from carriers, while FAA's data comes from air traffic control. Further, FAA is only required to track delays attributed to the NAS when they are 15 minutes or greater,[32] whereas BTS generally tracks NAS delays for all flights, regardless of cause, that are delayed by 15 minutes or greater. As a result, many of the flights with NAS delays of fewer than 15 minutes could be classified as a no-match by FAA. However, while this variance accounts for many of the no-matches between FAA's and BTS' data, our analysis found that it does not account for most of the flights with NAS delays of fewer than 15 minutes where FAA tracks the NAS delay and is able to provide delay causes other than no-match. The high rate of disagreements between airline-reported NAS delays with FAA's NAS delay data

[31] Flights that meet BTS' reporting threshold—i.e., flights that experience arrival delays of 15 minutes or longer—can report delay minutes under multiple causal categories. As a result, flights that reported a NAS delay could have also reported delays under other causal categories such as air carrier or extreme weather.
[32] OIG's review of FAA's data found that between 2018 and 2022, FAA tracked NAS delay data for 61 percent of the flights that experienced a NAS delay that was shorter than 15 minutes.

suggests that airlines may face challenges in determining how to properly categorize these delays and may need better guidance.

While BTS is aware of the no-matches data and has access to it, the Agency has not taken any steps to analyze this data to understand why these inconsistencies exist and how to best address them. According to the BTS officials, they have not undertaken this analysis because they are not required to do so. Nevertheless, as noted earlier, the same Federal statute that established BTS states that the Director of BTS shall continually improve surveys and data collection methods to improve the accuracy and utility of transportation statistics. DOT's Information Dissemination Quality Guidelines further emphasize this requirement.

According to a senior OAI official, data users have not raised the consistency of the data as a problem. However, according to another BTS official, the Agency could use data visualization tools[33] to analyze the reported causes data. According to this official, a data visualization tool could help compare flight activity between city pairs, and if those flights experienced any delays or cancellations, the tool could analyze the causes of delays and identify any inconsistencies. BTS could then follow up with the carriers to evaluate their justification and suggest corrections as necessary. To date, BTS does not have plans to implement these actions, and the Agency has not developed processes or tools to assess the consistency of the data that carriers are submitting.

As noted earlier, DOT's OACP uses BTS' flight delay and cancellation data as one of several tools to monitor and analyze airline operating and scheduling performance. Additionally, when OACP staff identify a potential consumer protection violation, they will often refer to BTS causal data as one source of information to help develop an understanding of the underlying causes of any delays or cancellations. When we informed the OACP officials about the inconsistencies in reporting, they acknowledged that those inconsistencies could impact BTS' reported causes data. Furthermore, the OACP officials agreed that improved consistency in BTS data could be helpful because it may enable OACP to analyze airline operating and scheduling performance and the underlying reported causes of flight delays and cancellations more efficiently. Overall, inconsistencies in BTS' data can impact DOT's understanding of the nature and causes of flight delays and cancellations as well as the effectiveness of the action it takes to improve air travel.

[33] Data visualization tools are software packages that can be used to reveal patterns for analysis in business intelligence, making the data more understandable.

Conclusion

Flight delays and cancellations are prevalent in the aviation industry. Collecting reliable delay and cancellation data is critical to understanding and addressing these events. BTS takes several steps to verify the accuracy of on-time performance data it receives from reporting carriers. However, the Agency has not yet formalized its processes or established procedures to ensure carriers are reporting complete and consistent delays and cancellation data. By improving consistency in reported cause data, DOT will be better positioned to evaluate carriers' operating and scheduling performance and to take actions to benefit airline customers or improve the efficiency of the NAS.

Recommendations

To improve the completeness and consistency of on-time performance and reported causes data, we recommend that the Director of the Bureau of Transportation Statistics (BTS):

1. Update and formalize the Agency's standard operating procedures for collecting, processing, and disseminating airline on-time performance data and make sure that these procedures are disseminated internally and available to all relevant staff.

2. Conduct an assessment of whether the BTS Data Administrator position and any other position should be designated as mission critical.

3. Develop succession plans for any mission-critical positions.

4. Develop and implement a process to verify that carriers are reporting on-time performance data for all flights.

5. Evaluate the Agency's technical directives to identify areas where it could provide clarifying or additional guidance to promote consistency in the data reported by carriers.

6. Work with FAA to understand no-matches in the National Airspace System (NAS) delay data between BTS' Airline Service Quality Performance (ASQP) and FAA's Aviation System Performance Metrics (ASPM) databases; and improve the consistency of air carrier-reported NAS delay data by providing additional carrier guidance as applicable.

7. Develop and implement a process to assess the consistency across air carriers' reporting of causes of delay data.

Agency Comments and OIG Response

We provided BTS with our draft report on August 28, 2024, and received its response on September 27, 2024, which is included as an appendix to this report. BTS concurred with our seven recommendations and provided appropriate actions and completion dates. Accordingly, we consider all recommendations resolved but open pending completion of the planned actions.

Actions Required

We consider recommendations 1 through 7 resolved but open pending completion of planned actions.

Exhibit A. Scope and Methodology

This performance audit was conducted between February 2023 and August 2024. We conducted this audit in accordance with generally accepted Government auditing standards. Those standards require that we plan and perform the audit to obtain sufficient, appropriate evidence to provide a reasonable basis for our findings and conclusions based on our audit objectives. We believe that the evidence obtained provides a reasonable basis for our findings and conclusions based on our audit objectives.

Initially, our audit objectives were to (1) identify the reported causes of flight delays and cancellations and (2) assess the controls DOT has in place to ensure the completeness and accuracy of reported flight delay and cancellation data. However, due to the breadth of data collected by both FAA and BTS on flight delays and cancellations, in November 2023, we revised our audit objective to focus on the reliability of the flight delay and cancellation data collected by BTS. Specifically, our revised audit objective was to assess the controls BTS has in place to ensure the completeness and accuracy of on-time performance and reported causes data.

We reviewed applicable laws, regulations, and BTS policies and procedures, including BTS' Statistical Standards Manual, technical directives, and SOPs governing the Agency's collecting and processing of flight delay and cancellation data. We interviewed officials from BTS' OAI and Office of Transportation Analysis, FAA, and OACP. We also sampled 6 out of the 17 reporting carriers in calendar year 2022. Our sample was selected based on a simple random sample that was stratified by the type of reporting carrier: Legacy, Low-Cost Carrier and Ultra Low-Cost Carrier, and Regional. Our sample of six reporting carriers included two carriers from each stratum.

In our review of domestic scheduled air carrier flight delays and cancellations data, we downloaded and analyzed both BTS' and FAA's on-time performance data between 2018 and 2022. We downloaded BTS' data from the Agency's TranStats and FAA's data from its OPSNET databases. We matched the records between BTS' and FAA's data using the scheduled departure date, operating carrier, tail number, flight number, origin airport, destination airport, scheduled arrival and departure time, and scheduled block time. We generated the annual number of flights that were either delayed or canceled for the marketing carriers, and their codeshare carriers, that were required to report data to BTS.

We also used FAA's ASQP and BTS' on-time performance data to generate our no-match analysis for delayed flights. First, we kept those flights with an actual block time and grouped the records by the calendar year to generate the following variables: the annual number of scheduled flights; the annual number

of delayed flights; the annual number of delayed flights that cited the reported causes of delays either as carrier, weather, security, late-arriving aircraft, or NAS; and the annual number of delayed flights where FAA reported NAS as the cause of the delay but with different causes of OPSNET delays, such as weather, volume, runway, other, equipment, and including "No Match."[34] The same set of variables were generated separately using FAA versus BTS on-time performance data. Similarly, we grouped the records by the calendar year to generate the same on-time performance metrics for the canceled flights. We also determined the level of delayed and canceled flights and the percentage of no-match discrepancies for NAS-related delays and cancellations between the two databases.

We produced the percentage rate of no-matches for NAS delayed and canceled flights from 2018 to 2022. The rate of no-matches for delayed and canceled flights was calculated as the number of NAS-delayed flights that were classified as a no-match divided by the total number of NAS-delayed flights. We performed a similar analysis to determine the rate of canceled no-match flights.

[34] FAA defines no-match as the number of flights with a carrier-reported delay (in this case NAS-related) that was not matched by OPSNET data.

Exhibit B. Organizations Visited or Contacted

DOT

- Bureau of Transportation Statistics
  - Office of Airline Information
  - Office of Transportation Analysis
- Office of Aviation Consumer Protection

FAA

- Office of Performance Analysis
- David J. Hurley Air Traffic Control Command Center

Other Organizations

- Allegiant Air
- American Airlines
- Republic Airways
- SkyWest Airlines
- Southwest Airlines
- United Airlines
- MITRE Corporation
- Airlines for America

Exhibit C. List of Acronyms

ACARS	Aircraft Communication Addressing and Reporting System
ASPM	Aviation System Performance Metrics
ASQP	Airline Service Quality Performance
ATCR	Air Travel Consumer Reports
BTS	Bureau of Transportation Statistics
DOT	Department of Transportation
FAA	Federal Aviation Administration
FTE	full-time employees
NAS	National Airspace System
OACP	Office of Aviation Consumer Protection
OAG	Official Airline Guide
OAI	BTS' Office of Airline Information
OIG	Office of Inspector General
SOP	standard operating procedure

Exhibit D. Major Contributors to This Report

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Office of the Secretary
of Transportation

Bureau of
Transportation Statistics

MEMORANDUM

Date: September 27, 2024

To: Nelda Z. Smith

Assistant Inspector General for Aviation Audits

From: Patricia Hu PATRICIA SIE-TSENG HU Digitally signed by PATRICIA SIE-TSENG HU Date: 2024.09.27 10:06:16 -04'00'

Director, Bureau of Transportation Statistics

Subject: Management Response to Office of Inspector General (OIG) Draft Report: *The Bureau of Transportation Statistics Verifies the Accuracy of Flight Delay and Cancellation Data but Can Do More To Assess Its Completeness and Consistency*

The Bureau of Transportation Statistics (BTS), part of the Department of Transportation (DOT), is the preeminent source of statistics on commercial aviation. BTS analyzes systemwide and carrier-specific delay and cancellation data collected by its Office of Airline Information to understand the nature and causes of flight delays and cancellations. BTS strives to ensure the completeness and accuracy of data for on-time performance and reported causes of delays and cancellations by working with carriers to resolve inconsistencies with air traffic control data and publishing corrections when carriers discover problems with their submissions. Further, BTS works extensively with the Federal Aviation Administration to analyze and understand discrepancies between delays that are attributed to the National Airspace System verses delays reported by air carriers.

BTS continuously looks for opportunities to assess and adjust its processes to enhance product quality, manage risks, and improve efficiency of the data it publishes to the public. The OIG report correctly notes that BTS causal data on delays and cancellations is used by the Department's Office of Aviation Consumer Protection (OACP), a unit within the Office of the General Counsel. However, the statements in the report that inconsistencies in this data impacts OACP's understanding of the nature and causes of flight delays and cancellations and the effectiveness of the actions it takes to improve air travel are inaccurate. As noted elsewhere in the report, the BTS data is only one of several tools used to monitor and understand the nature and causes of flight delays and cancellations.

Upon our review of OIG's draft report, BTS concurs with the seven recommendations as written. We plan to implement recommendations 1 through 3 by May 31, 2025, and recommendations 4 through 7 by September 30, 2025.

We appreciated the opportunity to review the OIG Draft Report. Please contact Mr. Rolf Schmitt, BTS Deputy Director, at 202-366-9258 with any questions.

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